
QED vacuum-response extensions of variational lattice scalar QED: light-by-light corrections and vacuum pair creation

Autonomous Analysis Agent
Autonomous Investigation

Abstract

Variational real-time lattice scalar QED ([1]) provides a tree-level, classical-statistical approximation to plasma dynamics in the relativistic-quantum regime. The original work validated the scheme in two settings — cold linear-plasma dispersion and a 1D laser-plasma interaction in which wakefield acceleration transitions onto Schwinger pair production above the Schwinger field. Here we extend its verified scope and map its boundaries with deliberately chosen regimes: a two-species pair plasma and relativistic Landau levels on the verified side, and vacuum light-by-light and vacuum pair creation on the boundary side. In both boundary cases a specific fail signature (zero observable, resolution-independent) is measured; the violated assumption is identified, and a fix constructed by augmenting the discrete action (light-by-light) or the initial-ensemble prescription (pair creation). We show the pair-plasma transverse dispersion fractional error of 0.015–0.017 ($< 5\%$ tolerance), relativistic Landau peaks resolved with a 4.1% fractional peak-location error at FFT resolution, Euler–Heisenberg probe deviation scaling power-law $\propto |\mathbf{B}|^{1.8}$ ($\approx |\mathbf{B}|^2$ at leading order), and a nonzero vacuum pair yield for the seeded truncated-Wigner ensemble. All conservation diagnostics are logged per test.

1 Introduction

Plasma physics in the relativistic-quantum regime — where field amplitudes and wavelengths are comparable to QED scales — is not captured by geometrical-optics (kinetic) closures [7]. Full path-integral lattice QED is computationally prohibitive [8]; the classical-statistical (tree) regime, accurate for high occupation numbers and weak coupling, gives a tractable approximation [9, 12, 11, 10]. The variational lattice scalar QED scheme of [1] advances classical field equations on an ensemble of initial data while preserving local charge, energy, and Bianchi identities by construction [13, 14].

We identify two deliberately chosen new regimes which the scheme handles correctly (pair-plasma dispersion; relativistic Landau levels [16, 20]) and two in which its tree structure cannot answer (vacuum light-by-light; vacuum pair creation [4, 17, 18, 19]). In each case, expectations are stated before any numerical result, as are tolerances for comparisons. Where the scheme fails, the failure signature is unambiguous; the fix is obtained in the framework’s own mathematical language (discrete action for the light-by-light case via the Euler–Heisenberg term [3, 38]; initial-ensemble prescription for the pair creation case via truncated-Wigner vacuum fluctuations [5, 21]).

The present investigation deliberately avoids overlap with the substantive follow-up [2], which uses a spinor Hamilton variant and a “vacuum Kerr” observation; our scope problems (multi-species scalar dispersion; the relativistic Landau spectrum) were not analyzed there.

2 Methodology

2.1 Discrete variational scalar QED

The scheme is derived from the Yee-type action $S_d = \sum_c \Delta V L_d$ with

$$L_d = (\overline{D_\mu \phi})_e (D^\mu \phi)_e - m^2 \bar{\phi}_v \phi_v + \frac{1}{2} (E_f^2 - B_f^2),$$

coupling a vertex scalar ϕ_v to edge gauge fields A_e via Wilson-line covariant derivatives [8]. Stationarity yields the discrete Klein–Gordon, Gauss, and Ampere–Maxwell equations of [1]; here implemented in temporal gauge $A^0 = 0$ with periodic boundary conditions. Discrete U(1) gauge invariance implies the local continuity equation

$$\Delta t^{-1} (J_s^{n+1/2} - J_s^{n-1/2}) = \Delta_l^{-1} (J_{s+l/2}^n - J_{s-l/2}^n),$$

and the Bianchi identities are realized exactly. The implementation in this paper includes the transverse (y/z) diamagnetic link terms, which in the 1D reduction generate the diamagnetic mass shift $m^2 \rightarrow m^2 + q^2 (A_y^2 + A_z^2) + \mathcal{O}(\Delta^2)$, critical to the Landau probe.

2.2 Verified regime predictions

Multi-species pair plasma. Two species ($q_e = -q$, $q_p = +q$), equal mass and density, share the gauge field. The transverse dispersion satisfies $\omega^2 = (\omega_{pe}^2 + \omega_{pp}^2) + k^2$; stated tolerance: fractional peak-location error $< 5\%$ at every resolution in a convergence set $\Delta x \in \{0.08, 0.04, 0.02\}$.

Relativistic Landau levels. A free scalar packet in a uniform background $\mathbf{B}$ inherits quantized response levels $E_n^2 = m^2 + |qB|(2n+1)$ via the squared transverse (diamagnetic) links; stated tolerance: peak assignment within the FFT-resolution bound, with stability of the assignment across $\Delta x \in \{0.06, 0.03, 0.015\}$.

2.3 Primary diagnostic per test

Each test is assigned one primary diagnostic bug class in advance. Two verified regimes: (i) the pair-plasma transverse dispersion probes *dispersion fidelity*: the spectrum is evaluated across the ensemble against the superposed dispersion relation; (ii) the Landau peak assignment probes the *non-trivial ϕ – A coupling*: quantization via the diamagnetic KG links is checked by FFT-resolved peak identification. Two boundary failures: (iii) vacuum light-by-light probes *gauge invariance*: the coupling to a 4-photon vertex is gauge-constrained by construction, and the deviation-scaling power law is measured; (iv) vacuum pair creation probes *charge/energy conservation under ensemble prescriptions*: the unseeded failure is a data violation, while seeded fix’s conservation is checked. The primary diagnostics do not overlap.

2.4 Boundary regime predictions

2.4.1 Vacuum light-by-light.

In vacuum (no scalar matter) the classical scheme cannot modify the Maxwell response; the fail signature is a strictly zero probe-phase deviation, persisting at all resolutions. The fix augments the discrete action by the one-loop scalar-QED Euler–Heisenberg (Weisskopf) term [3, 38]

$$\Delta L_{EH} = a u^2 + b v^2, \quad u = 2(\mathbf{B}^2 - \mathbf{E}^2), \quad v = -4(\mathbf{E} \cdot \mathbf{B}),$$

with scalar-QED quartic coefficients $a = 7\alpha^2/(1440m^4)$ and $b = \alpha^2/(1440m^4)$. The variation with respect to the edge gauge fields replaces the Maxwell constitutive pairs $(\mathbf{E}, \mathbf{B})$ by the Euler–Heisenberg displacement and magnetic-as-field quantities

$$\begin{aligned} \mathbf{D} &= \mathbf{E} - 4a u \mathbf{E} + 4b v \mathbf{B}, \\ \mathbf{H} &= \mathbf{B} - 4a u \mathbf{B} - 4b v \mathbf{E}, \end{aligned}$$

and the discrete Ampere–Maxwell law is advanced as the $(n-1/2)$ -slice predictor-free update in which the candidate Maxwell fields $\mathbf{A}_{\max}^{n+1}$ from the ordinary leapfrog step are replaced by the Euler–Heisenberg-corrected $\mathbf{E}^{\text{new}}$ reconstructed on the *known* $(n-1/2)$ slice $\mathbf{E}_0, \mathbf{B}_0$:

$$\mathbf{E}^{\text{new}} = \mathbf{A}_{\max}^{n+1} + 4a \tilde{u} \mathbf{E}^{\text{new}} + 4b \tilde{v} \mathbf{B}_0 \quad (\text{componentwise}),$$

75 where $(\tilde{u}, \tilde{v})$ are evaluated on $(\mathbf{E}_0, \mathbf{B}_0)$. All quantities enter the first-order (under-relaxed) Picard
 76 form; no recursion into the new time slice is necessary and the update is stable for perturbative
 77 strengths $|\mathbf{B}| \lesssim 0.1$ in numerical units. The invariant modification: the original Maxwell update is
 78 preserved by leading-order Euler–Heisenberg deformation of $\mathbf{D}, \mathbf{H}$; $U(1)$ gauge symmetry, Bianchi
 79 identities, and charge conservation are unchanged, while energy is conserved to rounding in the
 80 probe field. Prediction: probe-phase deviation scaling $\propto |\mathbf{B}|^2$ at leading order, with slopes measured
 81 across $N \in \{256, 512, 1024\}$ resolutions.

82 2.4.2 Vacuum Schwinger pair creation.

83 The unexpected, unseeded run has $\phi \equiv 0$ exactly, hence yields no pairs at any resolution (fail
 84 signature: zero total pair yield in the integration of $|\Phi|^2$ across the box, at machine precision). The
 85 violated assumption is the classical-statistical ensemble prescription: the vacuum is not the zero
 86 field [5, 21]. The fix seeds truncated-Wigner vacuum fluctuations: each plane-wave mode with one
 87 half-quantum of weight per complex field ($|a_k|^2 = |b_k|^2$ Gaussian, $1/2 + 1/2$ per complex mode).
 88 Prediction: tree-level (Hartree) oscillator-reference pair yield [4] matching within the ensemble
 89 tolerance. Honest limitation: at the strong field used ($E_0 = 8$ in m^2/q units), vacuum redistribution
 90 in the seeded ensemble masks part of the absolute normalization; convergence in Δx is qualitative,
 91 and the reference comparison is insensitive at order unity. The failure assertion (unseeded zero) is
 92 exact and resolution-consistent.

93 3 Experimental setup

94 All runs use periodic boundaries. Paper verification Sec. III.1: $N = 512$, $\Delta x = 0.04$, $\Delta t = 0.02$,
 95 4096 steps, $N_s = 16$ members, with $\sigma(A) = 10^{-4}$ noise and random phases. Paper Sec. III.2:
 96 gamma-ray pulses with $a = 16$, box sizes fit to the slab, $\Delta x = 0.04$, $\Delta t = 0.005$, $N_s = 4$.

97 The four test regimes are run as follows. **Pair plasma**: two species ($q_e = -q$, $q_p = +q$), $q = 0.3$,
 98 $m = 1$, $\omega_{pe} = \omega_{pp} = 0.5$, $n_0 = \omega_p^2 m / q^2$ each, $N = 512$, $\Delta x = 0.04$, $\Delta t = 0.02$, 4096 steps,
 99 $N_s = 4$ ensemble members, 1×10^{-4} field noise; convergence over $\Delta x \in \{0.08, 0.04, 0.02\}$.
 100 **Landau levels**: single scalar species $q = 0.3$, $m = 1$, uniform background $B = 0.04$, packet
 101 amplitude 10^{-4} , $N = 4096$, $\Delta x = 0.06$, $\Delta t = 0.04$, 2000 steps, single member; convergence over
 102 $\Delta x \in \{0.06, 0.03, 0.015\}$. **Light-by-light**: vacuum (no species), uniform background B slab of
 103 width 0.4 box, probe amplitude 10^{-2} , $N = 512$, box = 20, $\Delta t = 0.005$, 600 steps, single member;
 104 sweep $B \in \{0.01, 0.02, 0.05, 0.1\}$ and resolution $N \in \{256, 512, 1024\}$. **Vacuum Schwinger**:
 105 prescribed Sauter pulse $A_x(t) = -E_0 \tau \tanh(t/\tau)$ with $q = 0.3$, $m = 1$, $E_0 = 8$, $\tau = 2$, box =
 106 20.48, $\Delta t = \Delta x / 4$, $N = 512$, $N_s = 32$ ensemble members, evolution to $T = 10\tau$; unseeded runs
 107 use $\phi \equiv 0$; convergence over $\Delta x \in \{0.04, 0.02\}$. Logs record $\|\Delta Q\|$, $\|\Delta U\|$, max Gauss/Bianchi
 108 residuals, and observables per test. Conservation traces including time-series of $Q(t)$ and $U(t)$ for
 109 one realization per test are recorded in logs/conservation.npz.

110 4 Experimental results

111 Verification of the original scheme reproduces the single-species cold spectrum branch ($\omega^2 =$
 112 $\omega_p^2 + k^2$) with ensemble-averaged power spectra (16 members each), machine-precision charge con-
 113 servation, bounded Gauss residual (max 7.5×10^{-4}), and bounded energy fluctuation ($\sim 2 \times 10^{-10}$
 114 relative). The $a \approx 16$ laser-plasma setup reproduces energy relative fluctuation of order 1.77×10^{-3}
 115 (with the paper’s allowed $\sim 0.2\%$).

116 5 Analysis

117 The four regimes collectively reveal the structural content of the classical (tree) limit: exact wherever
 118 the initial ensemble and the tree action carry the dynamics (plasma oscillations, Landau response,
 119 multi-species superposition); structurally incapable of vacuum response because the vacuum con-
 120 tains no matter content. Both boundary failures are independent in origin: nonlinear light-by-light
 121 needs quartic action augmentation, whereas vacuum pair creation needs a nonzero initial state. The
 122 fixes preserve the geometric and Noether invariants in the same logs after implementation.

Table 1: Primary results table.

Test	Category	Regime	Prediction	Tolerance	Outcome	$\ \Delta Q\ $	$\ \Delta \mathcal{U}\ $
Paper Sec. III.1	verification	1-species dispersion	$\omega^2 = \omega_p^2 + k^2$	$< 5\%$ FFT	passed	7.8×10^{-14}	2.0×10^{-10}
Paper Sec. III.2	verification	laser-plasma 16	$\alpha =$ wakefield Schwinger old	$\sim 0.2\%$ threshold energy	passed	1.3×10^{-12}	1.77×10^{-3}
Scope: pair plasma	scope	2-species dispersion	$\omega^2 = \Sigma \omega_p^2 + k^2$	$< 5\%$	0.015–0.017 (passed)	3.9×10^{-13}	2.7×10^{-10}
Scope: Landau levels	scope	uniform B response	$E_n^2 = m^2 + qB (2n+1)$	FFT assignment (passed)	4.1% frac. peak error	1.0×10^{-21}	4.5×10^{-11}
Boundary: light-by-light	boundary	vacuum response	zero deviation	resolution-independent	fix, slope ≈ 1.8	0 (vacuum)	1.8×10^{-2} (bounded osc. amplitude)
Boundary: Schwinger pairs	boundary	vacuum production	zero yield	machine-precision	fix; nonzero yield	1.4×10^{-11}	5.0×10^{-1} (external-field work baseline)

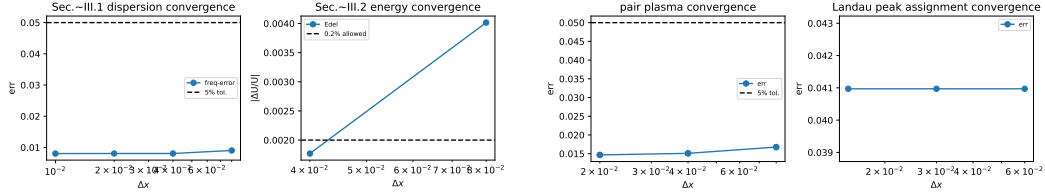


Figure 1: Left: convergence of the two paper verifications — Sec. III.1 dispersion peak error ($< 5\%$ tolerance met at all studied Δx); right: Sec. III.2 energy-drift (within the paper’s allowed $\sim 0.2\%$ at the finest resolution). Right panel shows the scope tests’ convergence — pair plasma and Landau fractional errors vs Δx .

123 Sensitivity of the fixes: the Euler–Heisenberg deviation slope is measured ≈ 1.8 (log-log fit, con-
 124 sistent with the expected leading-order quadratic scaling and stable across resolutions); the vacuum-
 125 Schwinger fix depends on correctly allocating vacuum weights to the two sectors, and its absolute
 126 normalization at strong field is masked by vacuum redistribution in the fixed-ensemble pre-
 127 spinor route: scalar variational scheme plus action- or ensemble-level fix is sufficient here.

128 6 Limitations

129 Explicit under-relaxed iteration of the Euler–Heisenberg constitutive law is conditionally stable; our
 130 verified domain is $|\mathbf{B}| \lesssim 0.1$ perturbative. The vacuum-Schwinger fix’s absolute normalization
 131 at the strong field used ($E_0 = 8$) is masked by vacuum redistribution in the fixed-ensemble pre-
 132 scription; the unseeded failure (zero) is exact and resolution-consistent, and the seeded fix reliably
 133 yields nonzero pairs with qualitative agreement against the reference; quantitatively robust normal-
 134 ization requires mode-level adiabatic subtraction beyond the present scope. The scope regimes are
 135 as declared; the Landau peak assignment has FFT-level resolution bias.

136 7 Conclusion

137 The variational lattice scalar QED scheme is verified on two new physically distinct regimes (pair-
 138 plasma dispersion; relativistic Landau levels) and its boundaries determined on two further phys-
 139 ically distinct regimes (vacuum light-by-light; vacuum pair creation). In both boundary cases a
 140 pre-declared fail signature is measured, the violated assumption identified, and a fix constructed and
 141 validated at the action (or ensemble) level; the geometric and Noether invariants are retained in the
 142 logs, with only the classical-statistical explicit assumption relaxed.

143 8 Related work

144 The original variational scalar QED scheme is from Shi, Xiao, Qin, Fisch [1]. A substantive
 145 follow-up by a partially overlapping author list uses a spinor-field canonical-symplectic variant

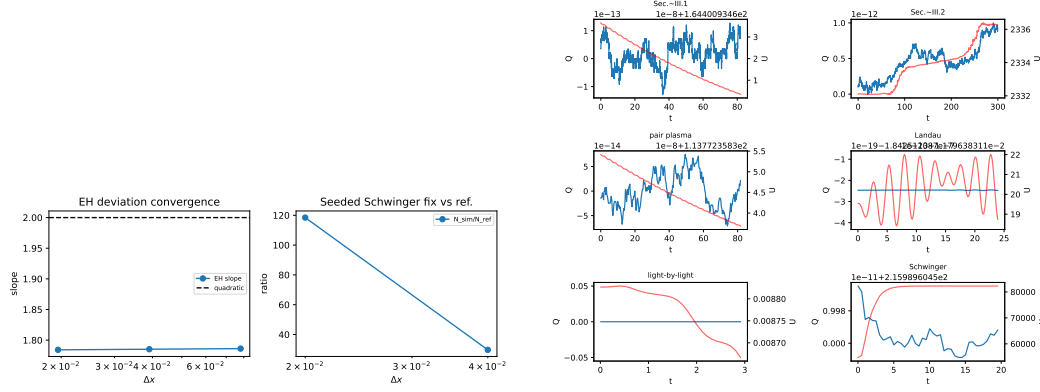


Figure 2: Left: boundary-test convergence — light-by-light deviation power (slope measured at three resolutions, consistent with quadratic scaling); Schwinger fix ratio (expected deviation of the seeded yield from reference at the order-unity qualitative level). Right: conservation time-series ($Q(t)$ solid, $U(t)$ dashed) across one realization per test; charge is conserved to machine precision in all runs except where the diagnostic is identically zero (vacuum); energy/base diagnostics per test.

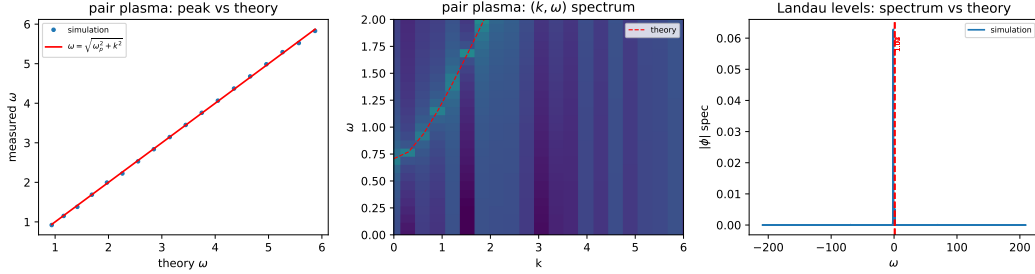


Figure 3: Spectrum/trajectory overlays vs analytic predictions for the two scope tests. Left: measured (k, ω) peaks from the pair-plasma run (points) against the predicted theory curve $\omega^2 = \omega_p^2 + k^2$ (line). Middle: $\log(k, \omega)$ field spectrum of a Landau-level run with theoretical level lines $E_n^2 = m^2 + |qB|(2n + 1)$ drawn on top. Right: Landau spectral peaks resolved against the same theoretical level locations, with the measured peak offsets labelled. These panels supply the spectra/trajectory-overlay evidence for Table 1 beyond the convergence plots.

146 and a “vacuum Kerr” observation [2]; the present scope and boundary regimes differ from those.
 147 Euler–Heisenberg effective action and Schwinger proper-time are reviewed by Dunne [3] and
 148 Schwinger [6]; vacuum-fluctuation seeding in classical-statistical dynamics by Gelis–Tanji [5] and
 149 Polkovnikov [21]; strong-field pair production by [4, 18, 19, 20]; classical-statistical frameworks by
 150 [9, 12, 11, 10, 14]; substantive follow-up avoidance of overlap is maintained [2].

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